.Model small -> CS and DS are.different and they are max 64KB

Dollar sign ($) -> string terminator

Anything written after

.STACK -> SS

.DATA -> DS

.CODE -> CS

String terminator is also stored in memory.

dw -> "CSE341$"

CS E3 41 $0 strings are stored in pairs in the memory with 0 if last one if odd number of chars.

Multiplication

Byte Multiplication:

One operand -> Must **AL***

Other operand -> other register

Format: **MUL other reg (no const.)**

Product is saved in **AX** register as maximum bits can be 16 if two 8 bits are multiplied.

Word Multiplication:

One operand -> Must AX*

Other operand -> other register

Format: **MUL other reg (no const.)**

Product is saved in **DX:AX** register as maximum bits can be 32 if two 8 bits are multiplied.

Division: divisor determines the type of division

Byte Division:

Dividend -> **AX** (16 bits)

Divisor -> other register (8 bits)

Format: **DIV other reg/divisor (no const.)**

Quotient and remainder can be maximum of 8 bit and they are saved in **AL and AH.**

Lower (AL) -> Quotient

Higher (AH) -> Remainder

Word division:

Dividend -> **DX:AX** (32 bits)

Divisor -> other register (16 bits)

Format: DIV other reg/divisor (no const.)

Quotient and remainder can be maximum of 16 bit and they are saved in **AX and DX.**

Lower (AX) -> Quotient

Higher (DX) -> Remainder